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POCKETED BUCKLING MEMBER ASSEMBLY

Abstract

A pocketed buckling member assembly comprises a plurality of parallel strings of pocketed buckling members. Each string is joined to at least one adjacent string. Each string comprising one piece of fabric folded around the buckling members into first and second opposed plies. Pockets are formed along each string by transverse seams joining the first and second plies. At least one buckling member is positioned in each pocket. The buckling member may have additives therein.

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Background/Summary

FIELD OF THE INVENTION

[0001] This invention relates generally to bedding and seating products and, more particularly, to pocketed buckling member assemblies used in bedding and seating products.

BACKGROUND OF THE INVENTION

[0002] Mattress spring core construction over the years has been a continuously improving art with advancements in materials and machine technology. A well-known form of spring core construction is known as a Marshall spring construction wherein metal coil springs are encapsulated in individual pockets of fabric and formed as elongate or continuous strings of pocketed coil springs. In an earlier form, these strings of coil springs were manufactured by folding an elongate piece of fabric in half lengthwise to form two plies of fabric and stitching transverse and longitudinal seams to join the plies of fabric to define pockets within which the springs were enveloped.

[0003] Improvements in spring core constructions have involved the use of fabrics which are thermally or ultrasonically weldable to themselves. By using such welding techniques, these fabrics have been advantageously used to create strings of individually pocketed coil springs wherein transverse and longitudinal welds, instead of stitching, are used to form the pockets encapsulating the springs.

[0004] Once strings of pocketed springs are constructed, they may be assembled to form a pocketed spring core or assembly for a mattress, cushion or the like by a variety of methods. For example, multiple or continuous strings may be arranged in a row pattern corresponding to the desired size and shape of a mattress or the like, and adjacent rows of strings may be interconnected by a variety of methods. The result is a unitary assembly of pocketed coil springs serving as a complete spring core assembly.

[0005] Conventional pocketed spring cores incorporating pocketed strings of springs have less motion transfer between sleeping partners when compared to traditional helically-laced open coil spring assemblies. Each pocketed coil spring moves with greater independence and, therefore provide less influence on adjacent pocketed coil springs than if the coil springs were not inside individual pockets. However, with a traditional pocketed spring mattress, a sheet of foam or other cushioning layer is attached to an upper surface of the pocketed spring assembly. The foam or cushioning sheet or sheets acts like a bridge, such that a load applied to one side of a mattress affects the other side of the mattress, providing an undesirable bridging effect. The present invention eliminates the undesirable bridging effect by encapsulating individual cushion members inside outer pockets of strings of springs.

[0006] U.S. Pat. No. 11,229,299 discloses a pocketed spring assembly incorporating strings which have individually pocketed coil springs with a cushioning pad and a buckling member inside the outer pockets. However, due to the inner coil springs being individually pocketed and contained within an outer pocket with a cushioning pad and a buckling member, the manufacture of such a product may be expensive.

[0007] Therefore, there remains a need to combine multiple technologies to reduce the manufacturing costs of a bedding or seating product having buckling members inside pockets.

[0008] There is further a need for a buckling member having additives to provide desired properties to a pocketed buckling member layer for use in a bedding or seating product.

SUMMARY OF THE INVENTION

[0009] In one aspect, a bedding or seating product is provided. The product comprises a pocketed buckling member assembly comprising a plurality of parallel strings of pocketed buckling members, each string being joined to at least one adjacent string. Each string comprises a piece of fabric surrounding a plurality of buckling members, first and second opposed plies of fabric being on opposite sides of the buckling members. A plurality of pockets having the same height are formed along the length of the string by a longitudinal seam and transverse seams joining the first and second plies. A single buckling member is in each of the pockets. Each buckling member has a height approximately equal to the height of the pocket in which the buckling member resides. Each of the buckling members may be a unitary member having an outer wall, a center, and internal ribs extending between the outer wall and center. These buckling members may have an open top and open bottom which define passages between the internal ribs. In some embodiments, each of the buckling members may have at least three internal ribs. The buckling members may have any number of passages of any desired size or shape.

[0010] Each of the buckling members is made of one of the following: acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, ethylene vinyl acetate (EVM) rubber, halogenated nitrile rubber, isoprene rubber, natural rubber, neoprene, polythiolethers, silicone rubber, styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene, polypropylene and blends thereof.

[0011] Cushioning materials may be placed on the pocketed buckling member assembly, and a covering, usually an upholstered covering, encases the pocketed buckling member assembly and cushioning materials.

[0012] The strings may extend longitudinally (from end-to-end) or transversely (from side-to-side). A pocketed buckling member assembly for use in a bedding or seating product may be posturized into regions or zones of different firmness by incorporating different strings into the pocketed buckling member assembly.

[0013] At least some of the buckling members may contain at least one of the following additives: phase change materials, plant-based fibers, synthetic fibers, fragrances, slow-release fragrances, organic and inorganic antimicrobial additives, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes, density reducing agents or gas generating materials. Such additives may impart different properties to the buckling members such as firmness, for example. The additives may be in the form of conductive solids, powders, particles or flakes.

[0014] At least some of the buckling members may have an outer wall and a hollow interior without any internal ribs or center. These buckling members may have an open top and open bottom which define one passage.

[0015] In another aspect, a pocketed buckling member assembly for a bedding or seating product is provided. The pocketed buckling member assembly comprises a plurality of parallel strings of pocketed buckling members. Each string is joined to at least one adjacent string. Each of the strings comprises a plurality of interconnected pockets made from one piece of fabric. Each of the pockets contains one buckling member encased in fabric, the buckling member filling the pocket with nothing else inside the pocket. The piece of fabric is joined to itself along a longitudinal seam and has first and second opposed plies of fabric on opposite sides of the buckling members. The first and second plies of fabric are joined by transverse seams between adjacent buckling members creating individual pockets, each pocket containing a single buckling member. Each of the buckling members is made of one of the following: acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, ethylene vinyl acetate (EVM) rubber, halogenated nitrile rubber, isoprene rubber, natural rubber, neoprene, polythiolethers, silicone rubber, styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene, polypropylene and blends thereof.

[0016] At least some of the buckling members may contain at least one of the following additives: phase change materials, plant-based fibers such as cellulose, cotton, hemp, linen, jute, wool and the like, synthetic fibers including carbon fiber, nylon, olefin, acrylic, polyester, rayon, vinyon, spandex, vinalon, aramids, Modal, PBI, Orlon and the like, fragrances including slow-release fragrances, organic and inorganic antimicrobial additives such as zinc omadine, N-Butyl-1,2-Benzisothiazonlin-3-one, OIT (2-n-Octyl-4-isothiazolin-3-one, 3-(Trimetholyxysily) proplydimethyloctadecyl ammonium chloride, copper and silver and the like, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes such as activated carbon, carbon nanotubes, graphene, graphite, aluminum powder, silicon carbide, diamond dust and the like may be incorporated into the material. Density reducing agents such as fumed silica or gas generating materials may be added too.

[0017] In some embodiments, each of the buckling members is a unitary member having an outer wall, internal ribs extending inwardly from the outer wall to a center, an open top and an open bottom defining passages between the internal ribs. In some of these embodiments, the buckling member has at least three internal ribs.

[0018] In another aspect, a pocketed buckling member assembly comprises a plurality of parallel strings of pocketed buckling members. Each string is joined to an adjacent string. Each of the strings comprises a plurality of interconnected outer pockets made from one piece of fabric. Each of the pockets is approximately the same height. Each pocket contains a single buckling member having the same height as the pocket. The piece of fabric is joined to itself along a longitudinal seam and has first and second opposed plies of fabric on opposite sides of the buckling members. The fabric of the first and second plies is joined by transverse seams. Some of the strings of pocketed buckling members have buckling members made of a different material than the buckling members of other strings. However, the buckling members within a string are the same or substantially the same. Such a pocketed buckling member assembly may have edge support due to the buckling members of the outermost strings being different than the buckling members of the interior strings.

[0019] Each of the buckling members may be a unitary member having a hollow interior and being made of one of the following: acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, ethylene vinyl acetate (EVM) rubber, halogenated nitrile rubber, isoprene rubber, natural rubber, neoprene, polythiolethers, silicone rubber, styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene, polypropylene and blends thereof.

[0020] At least some of the buckling members contain at least one of the following additives: phase change materials, plant-based fibers, synthetic fibers, fragrances, slow-release fragrances, organic and inorganic antimicrobial additives, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes, density reducing agents or gas generating materials.

[0021] In any embodiment shown or described herein, at least some of the buckling members may be a unitary member having an outer wall, a center, and internal ribs extending between the outer wall and center. These buckling members may have an open top and open bottom which define passages between the internal ribs. In some embodiments, each of the buckling members may have at least three internal ribs. The buckling members may have any number of passages of any desired size or shape.

[0022] One advantage of the present invention is that when a bedding or seating product, such as a mattress, is manufactured, the manufacturer need not purchase metal coil springs, which may be expensive.

[0023] Another advantage of the present invention is that the buckling members may be individually pocketed after having been formed with additives. The additives may impart a cooler “feel” to the user of the bedding or seating product in some areas or regions of a bedding or seating product when compared to other areas or regions.

[0024] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the summary of the invention given above, and the detailed description of the drawings given below, serve to explain the principles of the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. **1** is a perspective view, partially broken away, of a bedding or seating product incorporating a pocketed buckling member assembly surrounded by a border.

[0026] FIG. **1**A is a perspective view, partially broken away, of a bedding or seating product incorporating another pocketed buckling member assembly surrounded by a border.

[0027] FIG. **1**B is a perspective view, partially broken away, of a double-sided bedding product incorporating another pocketed buckling member assembly surrounded by a border.

[0028] FIG. **1**C is a perspective view, partially broken away, of another bedding product incorporating the pocketed buckling member assembly of FIG. **1** surrounded by a border.

[0029] FIG. **1**D is a perspective view, partially broken away, of another bedding product incorporating the pocketed buckling member assembly of FIG. **1**A.

[0030] FIG. **1**E is a perspective view, partially broken away, of another bedding product incorporating the pocketed buckling member assembly of FIG. **1**D surrounded by a foam border.

[0031] FIG. **2** is a perspective view, partially broken away, of a portion of a string of pocketed buckling members of the pocketed buckling member assembly of FIG. **1** in an unloaded condition.

[0032] FIG. **3** is a perspective view of a buckling member in a relaxed or unloaded condition.

[0033] FIG. **3**A is a perspective view of the circled area **3**A of FIG. **3** showing an additive incorporated into the buckling member.

[0034] FIG. **4** is a cross-sectional view, partially broken away, of a portion of the string of FIG. **2** in an unloaded condition.

[0035] FIG. **4**A is a cross-sectional view, partially broken away, of a portion of the string of FIG. **2** in an unloaded condition and a portion of a border comprising pocketed springs.

[0036] FIG. **4**B is a cross-sectional view, partially broken away, of a portion of the string of FIG. **2** in an unloaded condition and a portion of a border comprising pocketed buckling members.

[0037] FIG. **5** is a perspective view of a portion of the pocketed buckling member assembly of FIG. **1** in a relaxed condition.

[0038] FIG. **6** is a perspective view of a portion of another pocketed buckling member assembly in a relaxed condition, the strings being offset from one another.

[0039] FIG. **7**A is a top view of an alternative buckling member.

[0040] FIG. **7**B is a top view of an alternative buckling member.

[0041] FIG. **7**C is a top view of an alternative buckling member.

[0042] FIG. **7**D is a top view of an alternative buckling member.

[0043] FIG. **7**E is a top view of an alternative buckling member.

[0044] FIG. **7**F is a top view of an alternative buckling member.

[0045] FIG. **7**G is a top view of an alternative buckling member.

[0046] FIG. **7**H is a top view of an alternative buckling member.

[0047] FIG. **7**I is a top view of an alternative buckling member.

[0048] FIG. **7**J is a top view of an alternative buckling member.

[0049] FIG. **7**K is a top view of an alternative buckling member.

[0050] FIG. **7**L is a top view of an alternative buckling member.

[0051] FIG. **7**M is a top view of an alternative buckling member.

[0052] FIG. **7**N is a top view of an alternative buckling member.

[0053] FIG. **8** is a top view of a posturized pocketed spring assembly.

[0054] FIG. **9** is a top view of another posturized pocketed spring assembly.

[0055] FIG. **10** is a perspective view, partially broken away, of another bedding product incorporating a foam core and having a topper comprising a mini pocketed buckling member assembly.

[0056] FIG. **10**A is a perspective view, partially broken away, of a portion of one of the strings of the topper of FIG. **10**.

[0057] FIG. **10**B is a perspective view of a buckling member used in the topper of FIG. **10** in a relaxed or unloaded condition.
[0058] FIG. **11** is a perspective view, partially broken away, of a bedding product like FIG. **10** but having a posturized topper comprising a mini pocketed buckling member assembly.
[0059] FIG. **12** is a cross-sectional view, partially broken away, of another bedding product incorporating a foam core and having a topper comprising a pocketed buckling member assembly and foam topper pieces.
[0060] FIG. **13** is an enlarged perspective view of a portion of a comfort layer partially disassembled and a showing a portion of a welding tool.
[0061] FIG. **14** is an enlarged perspective view of a portion of another comfort layer partially disassembled and a showing a portion of another welding tool.

DETAILED DESCRIPTION OF THE INVENTION

[0062] Referring first to FIG. **1**, there is illustrated a bedding product in the form of a single-sided mattress **10** incorporating the principles of the present invention. This product or mattress **10** comprises a pocketed buckling member assembly **12** over the top of which there lay conventional padding or cushioning layers **14**, **16** which may be foam, fiber, gel, a pocketed spring blanket or any other suitable materials or any combination thereof. The pocketed buckling member assembly **12** is mounted upon a base **18** and is completely enclosed within an upholstered covering **20**. The base **18** may be any known material including a flexible sheet such as a scrim sheet or a more rigid dimensionally stabilizing substrate as described in U.S. Pat. Nos. 11,013,340 or 11,771,235, which are each incorporated by reference herein.
[0063] The pocketed buckling member assembly **12** is surrounded by a border **17** comprising longitudinally extending border strings **5** of pocketed members **11** and transversely extending border strings **6** of pocketed members **11**. The pocketed members **11** of the border **17** may be of a different diameter (usually smaller) than the diameter of the pocketed buckling members **28** of the pocketed buckling member assembly **12** to increase firmness along at least two edges or sides of the product. The pocketed members **11** of the border **17** may be border coil springs **96** of a diameter “D”, as shown in FIG. **4**A, or border buckling members **98** of a diameter “D”, as shown in FIG. **4**B.
[0064] Although one type of border **17** is illustrated, the border **17** may assume other forms or shapes of any desired size, such as pocketed border buckling members **98** of different geometries, different compositions, different sizes and/or different properties than the pocketed buckling members **28** of any of the pocketed buckling member assemblies disclosed herein. Alternatively, the border **17** may be omitted in this embodiment or any embodiment described or shown herein.
[0065] As shown in FIG. **1**, fully assembled, the product **10** has a length “L” defined as the linear distance between opposed end surfaces **22** (only one being shown in FIG. **1**). Similarly, the assembled product **10** has a width “W” defined as the linear distance between opposed side surfaces **24** (only one being shown in FIG. **1**). In the product shown in FIG. **1**, the length is illustrated as being greater than the width. However, it is within the scope of the present invention that the length and width may be identical, as in a square product.
[0066] As shown in FIG. **1**, the pocketed buckling member assembly **12** is manufactured from multiple strings **26** of pocketed buckling members **28** joined together. Each string **26** of pocketed buckling members **28** extends longitudinally or from head-to-foot along the full length of the product **10**.
[0067] Although the strings **26** of the pocketed buckling member assembly **12** are illustrated as extending longitudinally or from head-to-foot in the pocketed buckling member assembly **12** of FIG. **1**, they may extend transversely or from side-to-side as shown in the pocketed buckling member assembly **12***a* shown in the product **10***a* shown in FIG. **1**A. The pocketed buckling member assembly **12***a* comprises multiple strings **26***a* of pocketed buckling members, identical to the strings **26**, but shorter in length.
[0068] FIG. **1**B illustrates a double-sided mattress **10***b* comprising a pocketed buckling member

assembly **12** surrounded with a border **17**. However, the mattress **10***b* of FIG. **1**B has conventional padding layers **14**, **16** above and below a pocketed buckling member assembly **12**. The pocketed buckling member assembly **12** comprises a plurality of strings **26** extending from head to foot or longitudinally. However, the transversely extending strings **26***a* of FIG. **1**A could be used in a double-sided product including a mattress.

[0069] FIG. **1**C illustrates a single-sided mattress **10***c* comprising a pocketed buckling member assembly **12** surrounded by a border **17**. However, the mattress **10***c* of FIG. **1**C has a pocketed topper **19** comprising pocketed mini coil springs in addition to padding layers **14**, **16** above the pocketed topper **19**. A scrim layer **21** separates the pocketed topper **19** from the pocketed spring assembly **12**. Although one configuration of pocketed topper **19** is illustrated, any pocketed topper or comfort layer known in the art may be used.

[0070] FIG. **1**D illustrates a single-sided mattress **10***d* comprising pocketed buckling member assembly **12***a* the same as in the mattress **10***a* of FIG. **1**A. However, the mattress **10***d* of FIG. **1**D has no border. Although the strings **26***a* of pocketed buckling member assembly **12***a* extend transversely or from side to side, they may extend longitudinally or from head to foot without a border in any of the embodiments shown or described herein.

[0071] FIG. **1**E illustrates a single-sided mattress **10***e* comprising pocketed buckling member assembly **12** the same as in the mattress **10** of FIG. **1**. However, the mattress **10***e* of FIG. **1**E has a foam border **15**. Although the strings **26** of pocketed buckling member assembly **12** extend longitudinally or from head to foot, they may extend transversely or from side to side in any of the embodiments shown or described herein.

[0072] According to this invention, any of the padding or cushioning layers, including the pocketed topper **19**, may be omitted in any of the embodiments shown or described herein.

[0073] These strings of pocketed buckling members **26** and **26***a*, and any other strings described or shown herein, may be connected in side-by-side relationship as, for example, by gluing the sides of the strings together in an assembly machine, to create an assembly or matrix of springs having multiple rows and columns of pocketed buckling members bound together as by gluing, welding or any other conventional assembly process commonly used to create pocketed spring cores or assemblies.

[0074] Referring to FIGS. **5** and **6**, the strings **26** of pocketed buckling members may be joined so that the individually pocketed buckling members **28** are aligned in transversely extending rows **30** and longitudinally extending columns **32**. Alternatively, the strings **26** may be offset from one another in a pocketed buckling member assembly. In such an arrangement, shown in FIG. **6**, the individually pocketed buckling members **28** are not aligned in rows and columns; instead the individually pocketed buckling members **28** fill gaps or voids **70** of the adjacent strings **26**. FIG. **6** shows a portion of a pocketed spring assembly **12′** with multiple strings **26** arranged in this manner. Either alignment of strings may be incorporated into any of the pocketed buckling member assemblies or cores illustrated or described herein. Although FIGS. **5** and **6** illustrate strings **26**, the same alignments may be used in any pocketed buckling member assembly disclosed herein having any strings disclosed herein including strings **26***a*.

[0075] FIG. **2** illustrates a perspective view of the portion of a string **26** of pocketed buckling members **28** in a relaxed condition under no external load. As best illustrated in FIG. **2**, each string **26** of pocketed buckling members **28** comprises a row of interconnected fabric pockets **34**. Each of the fabric pockets **34** contains a single buckling member **36** and no other items. In other words, each of the fabric pockets **34** contains solely a single buckling member **36**.

[0076] Preferably, one piece of fabric is used to create the pockets **34** of the string **26** of pocketed buckling members **28**, the piece of fabric being folded over onto itself around the buckling members **28**. As best shown in FIG. **2**, opposite sides or plies **47**, **49** of the fabric are sewn, welded or otherwise secured together to create a longitudinal seam **50** and a plurality of separating or transverse seams **52**. FIG. **2** illustrates ply **47** being closest to the reader and ply **49** being behind the buckling members **28**.

[0077] Although the seams or welds in the embodiments shown herein are shown as being welded spaced rectangles, any of the seams may be spaced dots, triangles or solid line segments without spaces.

[0078] As best shown in FIG. **2**, opposed edges **56** of the piece of fabric used to create the string **26** of pocketed buckling members **28** are aligned and spaced from the longitudinal seam **50** a distance indicated by numeral **58**. Although the drawings indicated the longitudinal seam **50** being below the free edges **56** of the piece of fabric, the longitudinal seam **50** may be above the free edges **56** of the piece of fabric.

[0079] As shown in FIGS. **2**, **4**A and **4**B, the piece of fabric used to create the string **26** has a plurality of upper ears **53** and a plurality of lower ears **83**. When the ears are collapsed, the string **26** has a generally planar top surface **60** in a top plane P**1** and a parallel generally planar bottom surface **62** in a bottom plane P**2**. The linear distance between the top and bottom surfaces **60**, **62** of the string **26** defines a height “H” of the string **26**. This linear distance further defines the height H of the pocketed spring assembly **12** because each of the strings **26** has the same height. However, it is within the scope of the present invention that different strings of springs of a pocketed spring assembly have different heights.

[0080] As shown in FIGS. **4**A and **4**B, in one embodiment, the transverse seams **52** of string **26** separating adjacent pockets extend from the top of an upper ear **53** of fabric to the bottom of a lower ear **83** of fabric. However, the transverse seams **52** may be any desired length.

[0081] An ultrasonically weldable fabric is used to create the string of pocketed buckling members **26**. The ultrasonically weldable fabric is permeable to airflow through the fabric itself due to the nature of the fabric. Air moves between adjacent fabric pockets **38** and into and out of the string **26** through the ultrasonically weldable fabric.

[0082] As best shown in FIG. **2**, the longitudinal seam **52** comprises multiple spaced linear weld segments **64** formed using an ultrasonic welding horn and anvil (not shown) as known in the art. At least some of the longitudinal seams **52** of a string may not be segmented or be only partially segmented. For example, the longitudinal seam **52** of a string may not be segmented at all.

[0083] As best shown in FIGS. **4**A and **4**B, each transverse or separating seam **52** comprises multiple spaced linear weld segments **66** formed using an ultrasonic welding horn and anvil (not shown) to join the opposed sides or plies **47**, **49** of the ultrasonically weldable fabric. Again, at least some of the transverse or separating seams **52** of a string may not be segmented or may be only partially segmented. For example, one or more transverse seams **52** of a string may be partially segmented or not be segmented at all. Although the weld segments in the embodiments shown are illustrated as being heat-welded spaced rectangular-shaped segments, any of the seam segments may be other shapes, such as spaced dots, ovals or triangles of any desired sizes.

[0084] As best shown in FIGS. **2** and **3**, the buckling member **36** is shown as being cylindrically shaped member like a can. Buckling member **36** is illustrated having an outer wall **38** having an outer surface **40** and an inner surface **42**, the distance between which defines the thickness “T” of the outer wall **38**. The buckling member **36** has three internal ribs **44** which extend from the outer wall **38** to a center **46**. The buckling member **36** has a generally planar upper surface **67** and a generally planar lower surface **69** which define the height “HH” of the buckling member **36**. The buckling member **36** has an open top **71** and open bottom **73** defining three passages **75** between the internal ribs **44**. Although the drawings illustrate one thickness “T” of outer wall **38**, the outer wall may be any desired thickness. Although the drawings illustrate the internal ribs **44** having one thickness, the internal ribs may be any desired thickness.

[0085] In any embodiment shown or described herein, any of the buckling members **36**; **36***a*-**36***h*, **36***t* or **36***tt* shown or described herein can be made of acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, ethylene vinyl acetate (EVM) rubber, halogenated nitrile rubber, isoprene rubber, natural rubber, neoprene, polythiolethers, silicone rubber, styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene, polypropylene and blends thereof.

[0086] Regardless of the material used to make the buckling member **36**, additives **80** such as slow-release fragrances, antimicrobial additives such as copper and silver, color additives, thermochromic materials, bed bug inhibitors, conductive additives such as graphite, aluminum powder, silicon carbide and diamond dust may be incorporated into the material. Density reducing agents such as fumed silica or gas generating materials may be added too. FIG. **3**A illustrates additives **80** incorporated into the material of the buckling member **36**.

[0087] As shown in FIGS. **2** and **3**, the piece of fabric used to create the string **26** has a plurality of upper ears **53** and a plurality of lower ears **83**. When the ears are collapsed, the string **26** has a generally planar top surface **60** in a top plane P**1** and a parallel generally planar bottom surface **62** in a bottom plane P**2**. The linear distance between the top and bottom surfaces **60**, **62** of the string **26** defines a height "H" of the string **26**. This linear distance further defines the height H of the pocketed spring assembly **12** because each of the strings **26** has the same height. However, it is within the scope of the present invention that different strings of springs of a pocketed spring assembly have different heights.

[0088] As shown in FIGS. **2** and **3**, in one embodiment, the transverse seams **52** of string **26** separating adjacent pockets extend from the top of an upper ear **53** of fabric to the bottom of a lower ear **83** of fabric.

[0089] FIG. **4**A illustrates one embodiment of mattress **10** having a pocketed buckling member assembly **12** and border strings **6** of pocketed border coil springs **96** extending perpendicular to the head-to-foot or longitudinal direction of the strings **26** of buckling members **36**. Each of the border strings **6** extends from side-to-side or transversely. Each border string **6** comprises a single piece of fabric made into a plurality of individual border pockets **82** of the same height HHH defined as the linear distance between an upper surface **84** of the pocket in plane P**1** and a lower surface **86** of the border pocket **82** located in plane P**2**. The height HHH of the border strings **6** is preferably identical to the height H of the pockets **34** of the pocketed buckling member assembly **12**. However, these heights may be different. A coil spring **96** having a diameter D is located in each of the border pockets **82**. Although one configuration of coil spring **96** is illustrated any known coil spring may be used.

[0090] Although not shown, each of the border strings **5** extending from head-to-foot or longitudinally may contain a coil spring **96** and may be any desired height.

[0091] FIG. **4**B illustrates a view like FIG. **4**A but showing border buckling members **88** in place of coil springs **96** in the border pockets **82**. Each of the border buckling members **88** is shown having a diameter D, the same as the diameter of the coil springs **96**. However, the border buckling members **88** may have any desired diameter.

[0092] FIGS. **7**A-**7**H illustrate top views of different buckling members which may be substituted for the buckling members **36** shown and described herein. Although each of the buckling members shown in FIGS. **7**A-**7**H are illustrated being un-pocketed any of them may be individually pocketed. Regardless of the material used to make any of the buckling members, additives **80** such as phase change materials, plant-based fibers, synthetic fibers, fragrances, slow-release fragrances, organic and inorganic antimicrobial additives, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes, density reducing agents or gas generating materials may be incorporated into the material. Density reducing agents such as fumed silica or gas generating materials may be added too.

[0093] FIG. **7**A illustrates a cross-sectional view of another cylindrical shaped buckling member **36***a* which may be used in any embodiment of string shown or described herein. Buckling member **36***a* is illustrated having a circular outer wall **38***a* having an outer surface **40***a* and an inner surface **42***a*, the distance between which defines the thickness "Ta" of the circular outer wall **38***a*. The buckling member **36***a* has four internal ribs **44***a* which extend from the outer wall **38***a* to a center **46***a*. The buckling member **36***a* may be any desired height. The buckling member **36***a* has an open top and open bottom defining four internal passages **75***a* between the internal ribs **44***a* and outer wall **38***a*. Although the drawings illustrate one thickness "Ta" of outer wall **38***a*, the outer wall **38***a* may be any desired thickness. Although the drawings illustrate the internal ribs **44***a* having one thickness, the internal ribs may be any desired thickness.

[0094] FIG. **7**B illustrates a cross-sectional view of another cylindrical shaped buckling member **36***b* which may be used in any embodiment of string shown or described herein. Buckling member **36***b* is illustrated having a circular outer wall **38***b* having an outer surface **40***b* and an inner surface **42***b*, the distance between which defines the thickness "Tb" of the circular outer wall **38***b*. The buckling member **36***b* has no internal ribs and may be any desired height. The buckling member **36***b* has an open top and open bottom defining one internal passage **75***b* inside outer wall **38***b*. Although the drawings illustrate one thickness "Tb" of outer wall **38***b*, the outer wall **38***b* may be any desired thickness.

[0095] FIG. **7**C illustrates a cross-sectional view of another buckling member **36***c* which may be used in any embodiment of string shown or described herein. Buckling member **36***c* is illustrated having a triangular shaped outer wall **38***c* having an outer surface **40***c* and an inner surface **42***c*, the distance between which defines the thickness "Tc" of the triangular outer wall **38***c*. The buckling member **36***c* may be any desired height. The buckling member **36***c* has an open top and open bottom defining one internal passage **75***c* inside outer wall **38***c*. Although the drawings illustrate one thickness "Tc" of outer wall **38***c*, the outer wall **38***c* may be any desired thickness.

[0096] FIG. **7**D illustrates a cross-sectional view of another buckling member **36***d* which may be used in any embodiment of string shown or described herein. Buckling member **36***d* is illustrated having a square shaped outer wall **38***d* having an outer surface **40***d* and an inner surface **42***d*, the distance between which defines the thickness "Td" of the square outer wall **38***d*. The buckling member **36***d* may be any desired height. The buckling member **36***d* has an open top and open bottom defining one internal passage **75***d* inside outer wall **38**. Although the drawings illustrate one thickness "Td" of outer wall **38***d*, the outer wall **38***d* may be any desired thickness.

[0097] FIG. **7**E illustrates a cross-sectional view of another buckling member **36***e* which may be used in any embodiment of string shown or described herein. Buckling member **36***e* is illustrated having a pentagon shaped outer wall **38***e* having an outer surface **40***e* and an inner surface **42***e*, the distance between which defines the thickness "Te" of the pentagon outer wall **38***e*. The buckling member **36***e* may be any desired height. The buckling member **36***e* has an open top and open bottom defining one internal passage **75***e* inside outer wall **38***e*. Although the drawings illustrate one thickness "Te" of outer wall **38***e*, the outer wall **38***e* may be any desired thickness.

[0098] FIG. **7**F illustrates a cross-sectional view of another buckling member **36***f* which may be used in any embodiment of string shown or described herein. Buckling member **36***f* is illustrated having a hexagon shaped outer wall **38***f* having an outer surface **40***f* and an inner surface **42***f*, the distance between which defines the thickness "Tf" of the hexagon outer wall **38***f*. The buckling member **36***f* may be any desired height. The buckling member **36***f* has an open top and open bottom defining one internal passage **75***f* inside outer wall **38***f*. Although the drawings illustrate one thickness "Tf" of outer wall **38***f*, the outer wall **38***f* may be any desired thickness.

[0099] FIG. **7**G illustrates a cross-sectional view of another buckling member **36***g*which may be used in any embodiment of string shown or described herein. Buckling member **36***g* is illustrated having a heptagon shaped outer wall **38***g* having an outer surface **40***g* and an inner surface **42***g*, the distance between which defines the thickness "Tg" of the pentagon outer wall **38***g*. The buckling member **36***g* may be any desired height. The buckling member **36***g* has an open top and open bottom defining one internal passage **75***g* inside outer wall **38***g*. Although the drawings illustrate one thickness "Tg" of outer wall **38***g*, the outer wall **38***g* may be any desired thickness.

[0100] FIG. **7**H illustrates a cross-sectional view of another buckling member **36***h*which may be used in any embodiment of string shown or described herein. Buckling member **36***h* is illustrated having an octagon shaped outer wall **38***h* having an outer surface **40***h* and an inner surface **42***h*, the distance between which defines the thickness "Th" of the pentagon outer wall **38***h*. The buckling member **36***h* may be any desired height. The buckling member **36***h* has an open top and open bottom defining one internal passage **75***h* inside outer wall **38***h*. Although the drawings illustrate one thickness "Th" of outer wall **38***h*, the outer wall **38***h* may be any desired thickness.

[0101] FIG. **7**I illustrates a cross-sectional view of another buckling member **36***cc* which may be used in any embodiment of string shown or described herein. Buckling member **36***cc* is illustrated having the same configuration and structure as buckling member **36***c* but additionally has internal ribs **44***c*. Each of the internal ribs **44***c* extends from the outer wall **38***c* and converge at a center **46***c*. Although three internal ribs **44***c* are illustrated, any number of additional ribs may be added.

[0102] FIG. **7**J illustrates a cross-sectional view of another buckling member **36***dd* which may be used in any embodiment of string shown or described herein. Buckling member **36***dd* is illustrated having the same configuration and structure as buckling member **36***d* but additionally has internal ribs **44***d*. Each of the internal ribs **44***d* extends from the outer wall **38***d* and ends at a center **46***d*. Although four internal ribs **44***d* are illustrated, any number of additional ribs may be added.

[0103] FIG. **7**K illustrates a cross-sectional view of another buckling member **36***ee* which may be used in any embodiment of string shown or described herein. Buckling member **36***ee* is illustrated having the same configuration and structure as buckling member **36***e* but additionally has internal ribs **44***e*. Each of the internal ribs **44***e* extends from the outer wall **38***c* and ends at a center **46***e*. Although five internal ribs **44***e* are illustrated, any number of additional ribs may be added.

[0104] FIG. **7**L illustrates a cross-sectional view of another buckling member **36***ff* which may be used in any embodiment of string shown or described herein. Buckling member **36***ff* is illustrated having the same configuration and structure as buckling member **36***f* but additionally has internal ribs **44***f*. Each of the internal ribs **44***f* extends from the outer wall **38***f* and ends at a center **46***f*. Although five internal ribs **44***f* are illustrated, any number of additional ribs may be added.

[0105] FIG. **7**M illustrates a cross-sectional view of another buckling member **36***gg* which may be used in any embodiment of string shown or described herein. Buckling member **36***gg* is illustrated having the same configuration and structure as buckling member **36***g* but additionally has internal ribs **44***g*. Each of the internal ribs **44***g* extends from the outer wall **38***g* and ends at a center **46***g*. Although six internal ribs **44***g* are illustrated, any number of additional ribs may be added.

[0106] FIG. **7**N illustrates a cross-sectional view of another buckling member **36***hh* which may be used in any embodiment of string shown or described herein. Buckling member **36***hh* is illustrated having the same configuration and structure as buckling member **36***h* but additionally has internal ribs **44***h*. Each of the internal ribs **44***h* extends from the outer wall **38***h* and ends at a center **46***h*. Although eight internal ribs **44***h* are illustrated, any number of additional ribs may be added.

[0107] Referring now to FIG. **8**, longitudinally extending strings are shown in one preferable arrangement for a pocketed spring assembly for a bedding or seating product, such as a mattress. As can be seen, the longitudinally extending strings are arranged in a plurality of zones on the pocketed spring assembly **12***d*. By way of example, two zones **72**, **74** are illustrated, with the zones corresponding roughly to a "firm" side and a "soft" side. By way of further example, the longitudinally extending strings of the "soft" zone **72** may have a different buckling member **36** than the buckling members **36** of the longitudinally extending strings of the "firm" zone **74**. Of course, other arrangements are within the scope of the invention. For example, a pocketed spring assembly like pocketed spring assembly **12***d* shown in FIG. **8** may comprise transversely extending strings rather than longitudinally extending strings. In such an arrangement, each transversely extending string would have to be half firm and half soft. In one example, each string would have different buckling members.

[0108] Referring now to FIG. **9**, the transversely extending strings are shown in one preferable arrangement for a pocketed spring assembly **12***e* for a bedding or seating product, such as a mattress. As can be seen, the transversely extending strings are arranged in a plurality of zones on the pocketed spring assembly **12***e*. By way of example, three zones are illustrated, with the zones corresponding roughly to the location of a sleeper's head and shoulders, mid-section, knees and feet. By way of further example, the two end "soft" zones **90** each comprise strings of springs having a softer buckling members than the buckling members of the transversely extending strings of the middle or "firm" zone **92** are strings. Of course, other arrangements are within the scope of the invention. For example, the mattress shown in FIG. **9** may comprise longitudinally extending strings rather than transversely extending strings. In such an arrangement, each longitudinally extending string would have to be divided into three sections; a middle "firm" section and two end or "soft" sections. Therefore, each string would have only the end thirds of the string having softer buckling members than the middle third.

[0109] FIGS. **10-10**A illustrate an alternative embodiment of product such as a mattress **10***f* having a foam core **94** and a topper **112** residing on top of the foam core **94**. The topper **112** comprises a mini pocketed buckling member assembly **102** which has a bottom scrim sheet **100** which may be glued or otherwise adhered to an upper surface **99** of the foam core **94**. See FIG. **10**A. The mini pocketed buckling member assembly **102** of topper **112** is illustrated comprising multiple strings **26***f* of pocketed buckling members **28***f* which are shorter than the strings **26** of the pocketed buckling members **28** described herein. As seen in FIG. **10**, the strings **26***f* of pocketed buckling members **28***f* may be joined so that the individually pocketed buckling members **28***f* are aligned in transversely extending rows **30** and longitudinally extending columns **32**. Alternatively, the strings **26***f* may be offset from one another, as shown in FIG. **6**.

[0110] Although FIGS. **10-10**A illustrate a foam core **94**, the core on which the topper **112** resides may be any conventional core including a pocketed spring core, a pocketed buckling member core or combination of known elements.

[0111] The foam core **94** and a topper **112** residing above the foam core **94** are completely enclosed within an upholstered covering **20**. As shown in FIG. **10**, fully assembled, the mattress **10***f* has a length "L" defined as the linear distance between opposed end surfaces **22** (only one being shown in FIG. **10**). Similarly, the assembled mattress **10***f* has a width "W" defined as the linear distance between opposed side surfaces **24** (only one being shown in FIG. **10**). In the product shown in FIG. **10**, the length is illustrated as being greater than the width. However, it is within the scope of the present invention that the length and width may be identical, as in a square product.

[0112] FIG. **10**A illustrates a perspective view of the portion of a string **26***f* of pocketed buckling members **28***f* in a relaxed condition under no external load. As best illustrated in FIG. **10**A, each string **26***f* of pocketed buckling members **28***f* comprises a row of interconnected fabric pockets **34***f*. Each of the fabric pockets **34***f* contains a single buckling member **36***t*. For simplicity, like numbers are used for like parts.

[0113] The string **26***f* has a generally planar top surface **60***f* in a top plane P**1** and a parallel generally planar bottom surface **62***f* in a bottom plane P**2**. The linear distance between the top and bottom surfaces **60***f*, **62***f* of the string **26***f* defines a height "HT" of the string **26***f*. This linear distance further defines the height HT of the topper **112** because the strings **26***f* are approximately the same height. However, it is within the scope of the present invention that different strings of springs of the topper have different heights.

[0114] Although the topper **112** is shown comprising longitudinally extending strings **26***f* of individually pocketed buckling members **36***t*, the strings may extend transversely. Although the height HT of topper **112** is shown in the drawings to be a certain distance, the height HT of topper **112** may be any desired height but is preferably less than the height of the foam core **94**.

[0115] Although FIGS. **10-10**A illustrate buckling members **36***t* shaped like buckling members **36** but being a lesser height, the buckling members used in any topper shown or described herein may be any of the buckling members shown or described herein.

[0116] FIG. **10**B illustrates a buckling member **36***tt* shaped identical to buckling member **36***t* of

FIGS. **10-10**A and having the same height HT for use in a topper. However, buckling member **36***tt* has additives **80** mixed into the material used to make the buckling member **36***tt*. Regardless of the material used to make the buckling member **36***tt*, additives **80** such as slow-release fragrances, antimicrobial additives such as copper and silver, color additives, thermochromic materials, bed bug inhibitors, conductive additives such as graphite, aluminum powder, silicon carbide and diamond dust may be incorporated into the material. Density reducing agents such as fumed silica or gas generating materials may be added too.

[0117] If desired, buckling member **36***tt* with additives **80** may be used in place of any of the topper buckling members including buckling member **36***t* of FIGS. **10-10**A.

[0118] FIG. **11** illustrates an alternative embodiment of product such as a mattress **10***g* having a foam core **94** and a topper **110** comprising a topper middle section **106** and two topper end sections **108**. The topper **110** resides on top of the foam core **94**. Each topper end section **108** comprises a piece of foam located adjacent to the topper middle section **106** and may be secured to the topper middle section **106** using adhesive or any known method. The foam pieces **108** may be glued or otherwise adhered to an upper surface **99** of the foam core **94**. As shown in FIG. **11**, a conventional padding or cushioning layer **14** resides above the topper **110**.

[0119] The topper middle section **106** comprises a mini pocketed buckling member assembly **102** which has a bottom scrim sheet **100** which may be glued or otherwise adhered to an upper surface **99** of the foam core **94**. The mini pocketed buckling member assembly **102** of topper **110** is illustrated comprising multiple strings **26***g* of pocketed buckling members **28***g* which are shorter than the strings **26** of the pocketed buckling members **28** described herein. The mini pocketed buckling member assembly **102** of topper **110** has longitudinally extending columns **32** and transversely extending rows **30** of pocketed buckling members. Alternatively, the strings **26***g* may be offset from one another, as shown in FIG. **6**.

[0120] Although FIG. **11** illustrates a topper comprising three sections, the topper may comprise any number of sections, at least one of the sections comprising strings of pocketed buckling members. For example, the topper may have two sections oriented like “his” and “hers” sections. One of the topper sections may comprise strings of pocketed buckling members and the other topper section may comprise some other material.

[0121] Alternatively, in a topper being made solely of pocketed buckling members, some of the buckling members may have one or more additives as the term is defined herein while other buckling members have fewer or no additives. The different amount of additives may result in a posturized topper having different sections or areas of different firmness due to the different composition of the buckling members of the topper.

[0122] Referring to FIG. **11**, the foam core **94**, topper **110** and padding or cushioning layer **14** are completely enclosed within an upholstered covering **20**. As shown in FIG. **10**, fully assembled, the mattress **10***g* has a length “L” defined as the linear distance between opposed end surfaces **22** (only one being shown in FIG. **11**). Similarly, the assembled mattress **10***g* has a width “W” defined as the linear distance between opposed side surfaces **24** (only one being shown in FIG. **11**). In the product shown in FIG. **11**, the length is illustrated as being greater than the width. However, it is within the scope of the present invention that the length and width may be identical, as in a square product.

[0123] FIG. **12** illustrates an alternative embodiment of product such as a mattress **10***h* having a foam core **94** and a posturized topper **114** comprising different areas or regions of different firmness due to the buckling members inside the pockets of the posturized topper **114**. The posturized topper **114** resides on top of the foam core **94** and may be glued or otherwise adhered to an upper surface **99** of the foam core **94**. Although not shown, one or more conventional padding or cushioning layers may reside above the topper **114**. The posturized topper **114** comprises two end sections **132** and a middle section **134**, such a posturized topper may have any number of sections.

[0124] The mini pocketed buckling member assembly **128** of topper **114** is illustrated in FIG. **12** comprising multiple strings **26***h* of pocketed buckling members **28***h* which are shorter than the strings **26** of the pocketed buckling members **28** described herein. As seen in FIG. **12**, the strings **26***h* of pocketed buckling members **28***h* may be joined so that the individually pocketed buckling members **28***h* are aligned in transversely extending rows **30** and longitudinally extending columns **32**. Alternatively, the strings **26***h* may be offset from one another, as shown in FIG. **6**. Although the strings **26***h* are shown in FIG. **12** extending transversely or from side to side, they may be extend longitudinally.

[0125] Although FIGS. **11** and **12** illustrate a foam core **94** on which the toppers **110**, **114** reside, respectively, the core may be any conventional core including a pocketed spring core, pocketed buckling member core or combination of known elements.

[0126] FIG. **13** illustrates a pocketed comfort layer **116** made of two pieces of material, a first or upper ply **122** and a second or lower ply **124** which are welded together with circular weld seams **130**. The circular weld seams **130** comprise arcuate weld segments **126** with gaps **131** therebetween. Instead of mini coil springs as disclosed in U.S. Pat. Nos. 9,943,173; 9,968,202 and 10,813,462, each patent of which is fully incorporated by reference, mini buckling members **128** are located inside pockets **120** defined by the circular weld seams **130**.

[0127] The mini buckling members **128**, like the other buckling members described herein, may be made of any material described herein and may have any additives described herein. Similarly, mini buckling members **128**, like the other buckling members described herein, may be any desired shape or configuration described herein.

[0128] FIG. **14** illustrates a pocketed comfort layer **156** made of two pieces of material, a first or upper ply **122** and a second or lower ply **124** which are welded together with intersecting linear weld seams **170**. The linear weld seams **170** comprise straight weld segments **168** with gaps **177** therebetween. Instead of mini coil springs, mini buckling members **128** are located inside pockets **184** defined by the intersecting linear weld seams **170**.

[0129] Each of the buckling members shown or described herein may be made by at least one of the following methods: extrusion and cut to a desired member height or injection molding.

[0130] The various embodiments of the invention shown and described are merely for illustrative purposes only, as the drawings and the description are not intended to restrict or limit in any way the scope of the claims. Those skilled in the art will appreciate various changes, modifications, and improvements which can be made to the invention without departing from the spirit or scope thereof. The invention in its broader aspects is therefore not limited to the specific details and representative apparatus and methods shown and described. Departures may therefore be made from such details without departing from the spirit or scope of the general inventive concept. The invention resides in each individual feature described herein, alone, and in all combinations of any and all of those features. Accordingly, the scope of the invention shall be limited only by the following claims and their equivalents.

Claims

1. A bedding or seating product comprising: a pocketed buckling member assembly comprising a plurality of parallel strings of pocketed buckling members, each of said strings being joined to at least one adjacent string, each of said strings comprising a piece of fabric surrounding a plurality of buckling members, first and second opposed plies of fabric being on opposite sides of the buckling members, a plurality of pockets having the same height formed along said string by a longitudinal seam and transverse seams joining said first and second plies, a single buckling member being in each pocket and having a height approximately equal to the height of the pocket, wherein each of the buckling members is a unitary member having an outer wall, internal ribs extending inwardly from the outer wall to a center, an open top and an open bottom defining passages between the internal ribs, each of the buckling members being made of one of the following: acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene monomer

(EPDM) rubber, ethylene vinyl acetate (EVM) rubber, halogenated nitrile rubber, isoprene rubber, natural rubber, neoprene, polythiolethers, silicone rubber, styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene, polypropylene and blends thereof.

2. The product of claim 1 further comprising: cushioning materials; and a covering encasing said pocketed buckling member assembly and cushioning materials.

3. The product of claim 1, wherein at least some of the buckling members contain at least one of the following additives: phase change materials, plant-based fibers, synthetic fibers, fragrances, slow-release fragrances, organic and inorganic antimicrobial additives, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes, density reducing agents or gas generating materials.

4. The product of claim 1, wherein at least some of the buckling members have a different firmness than other buckling members.

5. The product of claim 4, wherein the pocketed buckling member assembly has a uniform height and the buckling member in each pocket has the same height as the pocketed buckling member assembly.

6. The product of claim 1, wherein at least some of the buckling members contain conductive additive solids, powders, particles or flakes.

7. The product of claim 1, wherein at least some of the buckling members have at least three internal ribs.

8. A pocketed buckling member assembly for a bedding or seating product, said pocketed buckling member assembly comprising: a plurality of parallel strings of pocketed buckling members, each of the strings being joined to at least one adjacent string, each of said strings comprising a plurality of innerconnected pockets, each of the pockets containing one buckling member encased in fabric, the buckling member filling the pocket, the fabric being connected to itself along a longitudinal seam and having first and second opposed plies of fabric on opposite sides of the buckling members, a plurality of pockets formed along said string by transverse seams joining said first and second plies, wherein each of the buckling members is a unitary member being made of one of the following: acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, ethylene vinyl acetate (EVM) rubber, halogenated nitrile rubber, isoprene rubber, natural rubber, neoprene, polythiolethers, silicone rubber, styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene, polypropylene and blends thereof.

9. The pocketed buckling member assembly of claim 8, wherein at least some of the buckling members contain at least one of the following additives: phase change materials, plant-based fibers, synthetic fibers, fragrances, slow-release fragrances, organic and inorganic antimicrobial additives, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes, density reducing agents or gas generating materials.

10. The pocketed buckling member assembly of claim 9, wherein at least some of the buckling members contain phase change materials.

11. The pocketed buckling member assembly of claim 8, wherein at least some of the buckling members contain conductive additive solids, powders, particles or flakes.

12. The pocketed buckling member assembly of claim 8, wherein at least some of the buckling members have an outer wall, internal ribs extending inwardly from the outer wall to a center, an open top and an open bottom defining passages between the internal ribs.

13. The pocketed buckling member assembly of claim 8, wherein the pocketed buckling member assembly has a uniform height and each buckling member in each pocket has the same height.

14. The pocketed buckling member assembly of claim 8, wherein at least some of the buckling members have an outer wall which is circular in cross-section.

15. A pocketed buckling member assembly for a bedding or seating product, said pocketed buckling member assembly comprising: a plurality of parallel strings of pocketed buckling members, each of the strings being joined to an adjacent string, each of the strings comprising a plurality of interconnected pockets made from fabric, each of the pockets being approximately the same height, each pocket containing a single buckling member having the same height as the pocket, the piece of fabric being joined to itself along a longitudinal seam and having first and second opposed plies of fabric on opposite sides of the buckling members, the fabric of said first and second plies being joined by transverse seams; some of the strings having buckling members made of a different material than the buckling members of other strings, and wherein each of the buckling members is a unitary member having hollow passages and being made of one of the following: acrylic rubber, acrylonitrile butadiene rubber, butadiene rubber, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, ethylene vinyl acetate (EVM) rubber, halogenated nitrile rubber, isoprene rubber, natural rubber, neoprene, polythiolethers, silicone rubber, styrene-butadiene, styrene copolymers, polystyrene, thermoplastic elastomers, vinyl methyl silicone, silicone, gel elastomer, polyurethane elastomer, polyurea elastomer, polyester, polyethylene and polypropylene.

16. The pocketed buckling member assembly of claim 15, wherein at least some of the buckling members contain at least one of the following additives: phase change materials, plant-based fibers, synthetic fibers, fragrances, slow-release fragrances, organic and inorganic antimicrobial additives, color additives, thermochromic materials, conductive additive solids, powders, particles, flakes, density reducing agents or gas generating materials.

17. The pocketed buckling member assembly of claim 15, wherein at least some of the buckling members have an outer wall, internal ribs extending inwardly from the outer wall to a center, an open top and an open bottom defining passages between the internal ribs.

18. The pocketed buckling member assembly of claim 15, wherein the buckling members are the same height.

19. The pocketed buckling member assembly of claim 15, wherein at least some of the buckling members contain conductive additive solids, powders, particles or flakes.

20. The pocketed buckling member assembly of claim 15, wherein at least some of the buckling members contain phase change materials.